Numerical Methods for Option Pricing

Stability, Convergence and Early Exercise

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Independent project
University of Surrey

June 2026

Abstract

This report prices European call and put options by four methods: the closed-form Black–Scholes formula, a Cox–Ross–Rubinstein binomial tree, Monte Carlo simulation, and an explicit finite-difference solver for the Black–Scholes equation, and extends the binomial method to American options. The four methods agree on a benchmark price to within the expected numerical error, which fixes the implementation. The closed-form price, being exact, then serves as a yardstick for studying the schemes: the binomial tree's oscillatory convergence, Monte Carlo's slow $1/\sqrt{n}$ error decay, and, for the finite-difference method, the conditional stability of the explicit scheme together with its second-order convergence in the space step. The American extension introduces the early-exercise decision, an optimal stopping problem with no closed-form solution, handled in the tree by a single comparison at each node.

Contents

1 Introduction

An option is a contract whose value depends on the uncertain future price of an underlying asset. Pricing it means assigning a fair value today, before that price is known. For the simplest contract under the simplest model, a European option on a stock following geometric Brownian motion, there is an exact answer, the Black–Scholes formula. Most contracts of practical interest have no such closed form and must be priced numerically.

This report takes the problem that *does* have an exact answer and prices it four different ways: the Black–Scholes formula itself, a binomial tree, Monte Carlo simulation, and a finite-difference solver for the Black–Scholes partial differential equation. Because the exact price is known, each numerical method can be checked against it, and its characteristic behaviour, how quickly it converges and when it breaks down, can be studied in a setting where the right answer is not in doubt. The same methods carry over to problems with no closed form; the controlled study here is what justifies trusting them there. The binomial method is then extended to American options, which can be exercised early and so have no closed-form price even under this model.

The scope is deliberately narrow. The underlying follows geometric Brownian motion with constant volatility and a constant risk-free rate, and the contracts are vanilla European and American calls and puts. The point is not to model market frictions but to study the numerical methods themselves against a known answer.

This project was completed independently during the second year of a BSc in Financial Mathematics at the University of Surrey, drawing in places on material beyond what has been formally covered in the degree at this point. Results that depend on that material, such as the derivation of the Black–Scholes equation itself, are cited to the standard references rather than re-derived; the contribution here is the implementation, cross-validation and numerical analysis of the four methods.

Section 2 sets up the model and the risk-neutral pricing rule that every method uses. Section 3 gives the Black–Scholes formula, which is the benchmark. Sections 4 to 6 develop the three numerical methods in turn, the last of these, the finite-difference scheme, being where stability and convergence are studied most closely. Section 7 treats American options and early exercise, and Section 8 draws the methods together and notes what each leaves for further work. All results are produced by a tested implementation and are reproducible.

2 The model and risk-neutral valuation

Every method in this report prices the same object, a European option on a single stock, and rests on the same two ingredients: a model for how the stock moves, and a rule for turning that model into a price. This section sets out both. The model is geometric Brownian motion; the pricing rule is risk-neutral valuation, which follows from the absence of arbitrage. The argument that produces the pricing rule also produces the Black–Scholes partial differential equation, which the finite-difference method later solves directly.

2.1 The underlying asset

We model the stock price S_t as geometric Brownian motion,

$$\mathrm{d}S_t = \mu S_t \,\mathrm{d}t + \sigma S_t \,\mathrm{d}W_t. \tag{2.1}$$

The equation has two parts. The first, $\mu S_t \,\mathrm{d}t$, is the *drift*: a steady trend, with μ the stock's expected rate of return. The second, $\sigma S_t \,\mathrm{d}W_t$, is the *shock*: a random fluctuation, with $\sigma > 0$ the volatility and W_t Brownian motion.

Both terms are scaled by S_t, for two reasons. First, it expresses a move as a proportion of the current price rather than a fixed number of pounds: a £1 change matters far more to a £10 stock than to a £100 stock, so scaling by S_t puts moves of different stocks, or of the same stock at different price levels, on a comparable footing. Second, it keeps the price positive: the model only ever multiplies S_t by something, never subtracts a fixed amount that could push it below zero, which a share price cannot do.

The random term $\mathrm{d}W_t$ is built from Brownian motion. Over a small time step $\mathrm{d}t$ it can be written

$$\mathrm{d}W_t = \varepsilon\sqrt{\mathrm{d}t}, \qquad \varepsilon \sim N(0,1),$$

a standard normal draw scaled by $\sqrt{\mathrm{d}t}$. The square-root scaling reflects how randomness accumulates: shocks over non-overlapping time intervals are independent, and variance adds across independent quantities, so the variance built up after a time t is proportional to t, and the standard deviation, the typical size of the accumulated move, is proportional to $\sqrt{t}$.

σ is quoted on an annual basis. Informally, under this model around two years out of three see an annual return within one standard deviation of its average, so $\sigma = 20\%$ corresponds to yearly moves of roughly $\pm 20\%$ being unremarkable rather than extreme.

2.2 The pricing problem

A European option is a contract paying a fixed function of S_T at a maturity T, and nothing before. A call with strike K pays

$$(S_T - K)^+ = \max(S_T - K,\, 0),$$

and a put pays $(K - S_T)^+$. The task is to value the contract at $t = 0$.

The obvious guess, discounting the expected payoff using your own forecast of the stock's chances, fails. It depends on μ, the stock's expected return, which different investors estimate differently; a price all of them can agree on cannot depend on it. The resolution, set out next, is to discount using a different, agreed set of odds instead of anyone's personal forecast.

2.3 Risk-neutral valuation

Write $V(S,t)$ for the value of the option at a future time t if the stock is at level S then. In particular $V(S_0,0)$ is today's price, with S_0 today's stock price: the quantity this report is ultimately after.

The stock's expected return μ and the risk-free rate r are different numbers. μ is the average return of the stock itself, a risky asset; r is the safe return available from something like a government bond. μ is normally higher than r, because investors need some extra reward for holding a risky asset rather than a safe one. Pricing the option using μ would mean pricing it using how large a reward each investor personally demands for that risk, which is not something everyone agrees on.

The standard fix is to price the option exactly as before, a discounted average payoff, but with μ replaced by r throughout the calculation:

$$V(S_0,0) = e^{-rT}\,\overline{\text{payoff}(S_T)}, \tag{2.2}$$

where the bar denotes the average payoff, computed assuming the stock grows at rate r rather than μ. That this substitution gives the correct price is a standard no-arbitrage result, established in the original Black–Scholes paper [1]. It is stated here rather than derived, because the result itself is what the report uses: with μ replaced by r, the price no longer depends on anyone's personal view of the stock. Section 4 carries out exactly this substitution by hand, for a single up-or-down step.

Under this substitution, the terminal stock price follows the same kind of formula as before with μ replaced by r,

$$S_T = S_0 \exp\!\left(\left(r - \tfrac{1}{2}\sigma^2\right)T + \sigma\sqrt{T}\,Z\right), \qquad Z \sim N(0,1). \tag{2.3}$$

This is the law sampled directly by the Monte Carlo method (section 5) and integrated in closed form by the Black–Scholes formula (section 3).

2.4 The Black–Scholes equation

Equation (2.2) has an equivalent description as a partial differential equation. Under the same model, the price $V(S,t)$ can be shown to satisfy the Black–Scholes equation,

$$\frac{\partial V}{\partial t} + \tfrac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0, \tag{2.4}$$

to be solved backwards from the payoff specified at $t = T$. This equation is derived in the original Black–Scholes paper [1]; it is stated here because section 6 solves it directly by finite differences. As with (2.2), μ does not appear: both describe the same risk-neutral price, one as an expectation and the other as a PDE, depending only on S, r and σ.

3 The Black–Scholes formula

For a European call or put, the expectation (2.2) over the lognormal terminal law (2.3) evaluates in closed form to the Black–Scholes formula [1]. For strike K and maturity T,

$$C = S\,\Phi(d_1) - Ke^{-rT}\Phi(d_2), \tag{3.1}$$

$$P = Ke^{-rT}\Phi(-d_2) - S\,\Phi(-d_1), \tag{3.2}$$

where Φ is the standard normal distribution function and

$$d_1 = \frac{\ln(S/K) + \left(r + \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}. \tag{3.3}$$

The two terms of the call (3.1) have a direct reading. The second, $Ke^{-rT}\Phi(d_2)$, is the strike, discounted to present value because it is paid at T, and weighted by $\Phi(d_2)$, the risk-neutral probability of exercise. The holder pays K only in the states where exercise happens, which is why the strike enters discounted and probability-weighted rather than as K. The first term is the corresponding present value of receiving the stock in those states.

The quantities d_1 and d_2 are not arbitrary. From section 2, the terminal price satisfies $\ln(S_T/S) \sim N((r - \frac{1}{2}\sigma^2)T,\ \sigma^2 T)$. The call finishes in the money when $S_T > K$, equivalently $\ln(S_T/S) > \ln(K/S)$; standardising this event against the known mean and standard deviation of $\ln(S_T/S)$ gives exactly d_2, so d_2 measures, in standard deviations, how far today's price sits above the strike once the risk-neutral drift is accounted for, and $\Phi(d_2)$ is the probability of that event. At the benchmark parameters $d_2 = 0.15$ and $\Phi(d_2) \approx 0.56$, a fractionally better than even chance the call finishes in the money. The other quantity, $d_1 = d_2 + \sigma\sqrt{T}$, is d_2 shifted by one unit of volatility, which weights the first term $S\,\Phi(d_1)$ correctly as the present value of the stock received on exercise. Unlike $\Phi(d_2)$, $\Phi(d_1)$ is not an ordinary probability: it is a truncated expectation, averaging S_T only over the scenarios where the option finishes in the money, so it already has how far above K baked in, as well as how likely. It also has a more concrete identity: $\Phi(d_1)$ is the option's delta, $\partial C/\partial S$, the amount the price moves for a £1 move in the stock. The reason for the shift between d_1 and d_2 involves a change of probability measure, a standard result of the risk-neutral pricing theory that is beyond this report's scope. At the benchmark parameters $d_1 = 0.35$ and $\Phi(d_1) \approx 0.64$.

A relation that does not depend on the model links the two. Subtracting (3.2) from (3.1),

$$C - P = S - Ke^{-rT}. \tag{3.4}$$

This is put–call parity, and it follows from the payoffs alone by a no-arbitrage argument: a portfolio long a call and short a put has the same payoff at T as holding the stock and borrowing Ke^{-rT}, so the two must cost the same today. It holds whatever prices a model assigns, so is a clean check on any pricer.

At the benchmark parameters used throughout ($S = K = 100$, $r = 5\%$, $\sigma = 20\%$, $T = 1$), formula (3.1) gives a call price of 10.4506. The implementation reproduces this, and put–call parity (3.4) holds to machine precision, both checked in the test suite: the reference for the three methods below.

4 The binomial tree

The binomial method replaces continuous time with a finite number of steps. Over each step the stock makes one of two moves, and the option is priced by working backwards from its payoff. As the number of steps grows, the price converges to the Black–Scholes value.

4.1 Construction

Divide $[0, T]$ into N steps of length $\Delta t = T/N$. Over one step the price either multiplies by $u > 1$ or by $d < 1$. The Cox–Ross–Rubinstein choice [2] sets

$$u = e^{\sigma\sqrt{\Delta t}}, \qquad d = e^{-\sigma\sqrt{\Delta t}} = \frac{1}{u}, \tag{4.1}$$

which makes the tree recombine: an up move followed by a down move returns to the starting price, so that after n steps there are only $n+1$ distinct prices rather than 2^n. The step is priced under the risk-neutral measure of section 2: requiring the expected one-step growth to equal the risk-free rate, $p\,u + (1-p)\,d = e^{r\Delta t}$, fixes the risk-neutral probability of an up move. Expanding the constraint, $p\,u + d - p\,d = e^{r\Delta t}$, so $p\,(u-d) = e^{r\Delta t} - d$, and dividing through by $(u-d)$ gives

$$p = \frac{e^{r\Delta t} - d}{u - d}. \tag{4.2}$$

Read as a probability, p is the chance of an up move and $1-p$ the chance of a down move; the two automatically sum to one, since $1-p$ simplifies to $(u - e^{r\Delta t})/(u-d)$, the same fraction with the numerator's two terms swapped. This p is the same kind of substitution introduced in section 2: it is not a forecast of which way the stock will actually move, but the value that makes the stock's expected one-step growth match the risk-free rate r, in place of the stock's real drift. Figure 1 shows the first few steps.

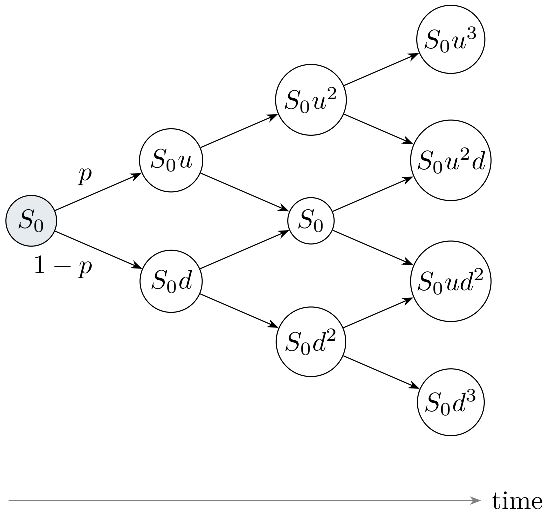

Figure 1: The first three steps of a recombining CRR tree. Each step moves up by a factor u with risk-neutral probability p or down by d with probability $1-p$; an up move and a down move recombine to the same price.

4.2 Pricing by backward induction

At maturity the value at each terminal node is just the payoff. Stepping back one level, a node's value is the discounted risk-neutral expectation of the two nodes it leads to,

$$V = e^{-r\Delta t}\big[\, p\, V_{\mathrm{up}} + (1-p)\, V_{\mathrm{down}} \,\big], \tag{4.3}$$

which is (2.2) applied over a single step. Repeating (4.3) from the terminal layer back to the root gives the price today.

4.3 Convergence

As $N \to \infty$ the tree price converges to the Black–Scholes value, but not smoothly: it does not approach 10.4506 steadily from one side. Figure 2 plots the price against N, and it wobbles, landing alternately a little too high and a little too low as N increases by one each time, with the wobble shrinking as N grows. To be precise about what is wobbling: each value of N defines one tree, and that whole tree collapses by backward induction to a single number, the price. It is this single number, one per N, that is plotted against N, not the scatter of terminal prices inside any one tree.

Table 1 makes the link between the two concrete, for $N = 2$ and $N = 3$ at the benchmark parameters ($S_0 = K = 100$, $\sigma = 20\%$, $T = 1$). The payoff $\max(S-K, 0)$ has a kink at $S = K$: it is flat below K and rises linearly above it. With $N = 2$, one of the three terminal prices lands exactly on K, because a recombining tree always has a path with equally many up and down moves when N is even, and an equal number of up and down moves returns the price to its starting level. With $N = 3$, no terminal price is anywhere near K, because an odd number of steps cannot be split into two equal halves; the nearest prices sit more than £11 away on either side.

	Terminal prices	Closest to $K = 100$	Resulting price
$N = 2$ (even)	75.4, **100.0**, 132.7	100.0, exact	9.54 (£0.91 below)
$N = 3$ (odd)	70.7, 89.1, 112.2, 141.4	over £11 away	11.04 (£0.59 above)

Table 1: Terminal tree prices for $N = 2$ and $N = 3$ at $S_0 = K = 100$, and the single price each whole tree produces. These are the first two points plotted in fig. 2.

The last column is the payoff for understanding the wobble: $N = 2$, with a node sitting exactly on the kink, undershoots the true price by £0.91; $N = 3$, with a gap straddling the kink, overshoots it by £0.59. A node on the kink lets that tree price the region around K more accurately than a tree with only a gap there, and this difference in accuracy is what pushes the computed price to one side of the true value or the other. The pattern holds generally: every even N undershoots, every odd N overshoots, and both gaps shrink steadily as N grows, the decaying zig-zag visible in fig. 2. The precise size of the under- and over-shoot at each N follows formally from how a discrete sum approximates a kinked function, treated in the original CRR paper [2]; node position relative to K is the mechanism demonstrated here.

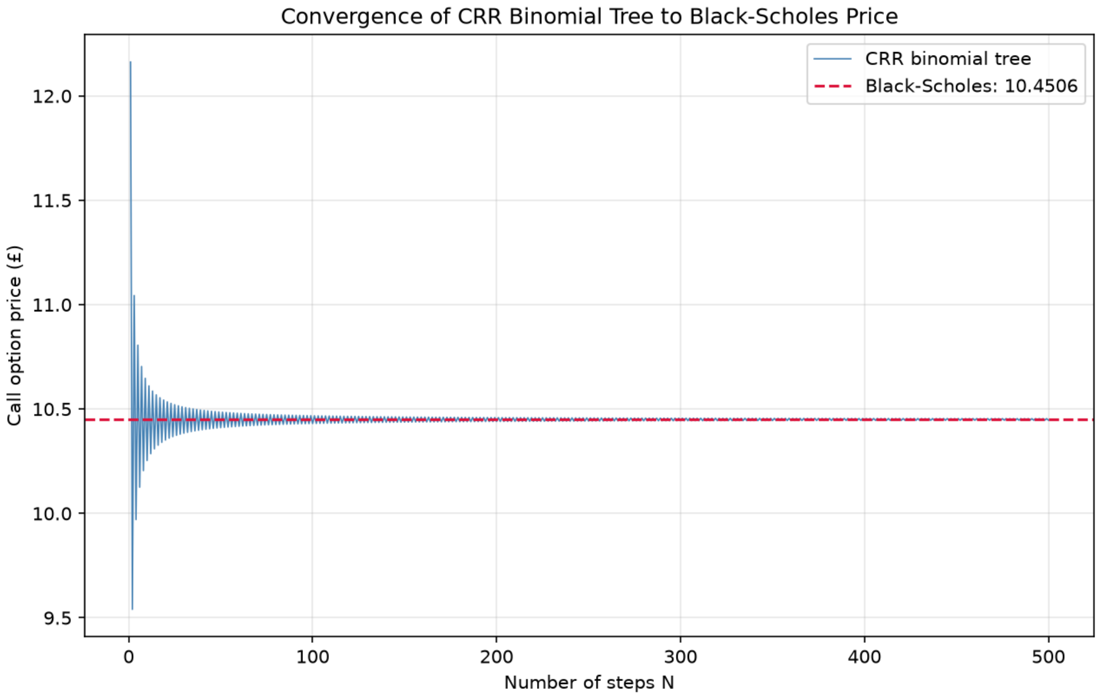

Figure 2: Binomial price against the number of steps N. The price converges to the Black–Scholes value with the characteristic odd–even oscillation, which decays as the step count grows.

At $N = 5000$ the tree gives 10.4502, matching the Black–Scholes price to three decimal places; the test suite checks agreement at large N. The method demonstrates a fully discrete scheme converging to the continuous-time price, with the oscillation a visible feature of how the discrete and continuous problems relate.

5 Monte Carlo simulation

Monte Carlo prices the option through the expectation form (2.2) directly: it estimates the discounted expected payoff by averaging over simulated outcomes.

5.1 The estimator

For a European payoff only the terminal price S_T matters, so it can be sampled directly from the lognormal law (2.3) without simulating the path in between. Drawing n independent standard normals $Z_1, \ldots, Z_n$ and setting $S_T^{(j)} = S_0 \exp\big((r - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T}\, Z_j\big)$ - the bracketed superscript (j) labels the j-th simulated price, not a power - the price is estimated by the discounted sample mean of the payoffs,

$$\widehat{V}_n = e^{-rT} \frac{1}{n} \sum_{j=1}^{n} \big(S_T^{(j)} - K\big)^+. \tag{5.1}$$

The estimator is unbiased: averaged over many repetitions, it returns exactly the Black–Scholes price.

5.2 Error and convergence

In plain terms: run the simulation once and the estimator returns one number; run it again with a fresh batch of random draws and it returns a slightly different number. The standard error measures how much these estimates typically vary from one batch to the next, and how that variation shrinks as more paths are simulated.

More precisely, by the central limit theorem $\widehat{V}_n$ is approximately normal about the true price with standard deviation $s/\sqrt{n}$, where s is the sample standard deviation of the discounted payoffs. This standard error sets the size of the estimate's likely error, and it shrinks only as $1/\sqrt{n}$. Because the n sits under a square root, halving the error needs $\sqrt{n}$ to double, which needs n itself to be multiplied by four: $\sqrt{4n} = 2\sqrt{n}$. The slow rate is the method's defining limitation, and the price of its generality: the same estimator works in any number of dimensions and for any payoff, where the lattice and the PDE become expensive or intractable. The simulation uses a fixed random seed, so every reported figure is reproducible.

Figure 3 shows the estimate settling onto the Black–Scholes price as n grows, with the swings around the limit shrinking as the standard error falls at the $1/\sqrt{n}$ rate. At $n = 10^6$ paths, the n discounted payoffs have a sample standard deviation of $s \approx 14.73$; dividing by $\sqrt{n} = 1000$ gives the standard error quoted earlier, $s/\sqrt{n} = 14.73/1000 = 0.0147$, so the estimate is 10.4532 ± 0.0147. This is consistent with the exact price 10.4506, lying within about two standard errors of it. The test suite checks that the estimate falls within a set tolerance of the Black–Scholes value.

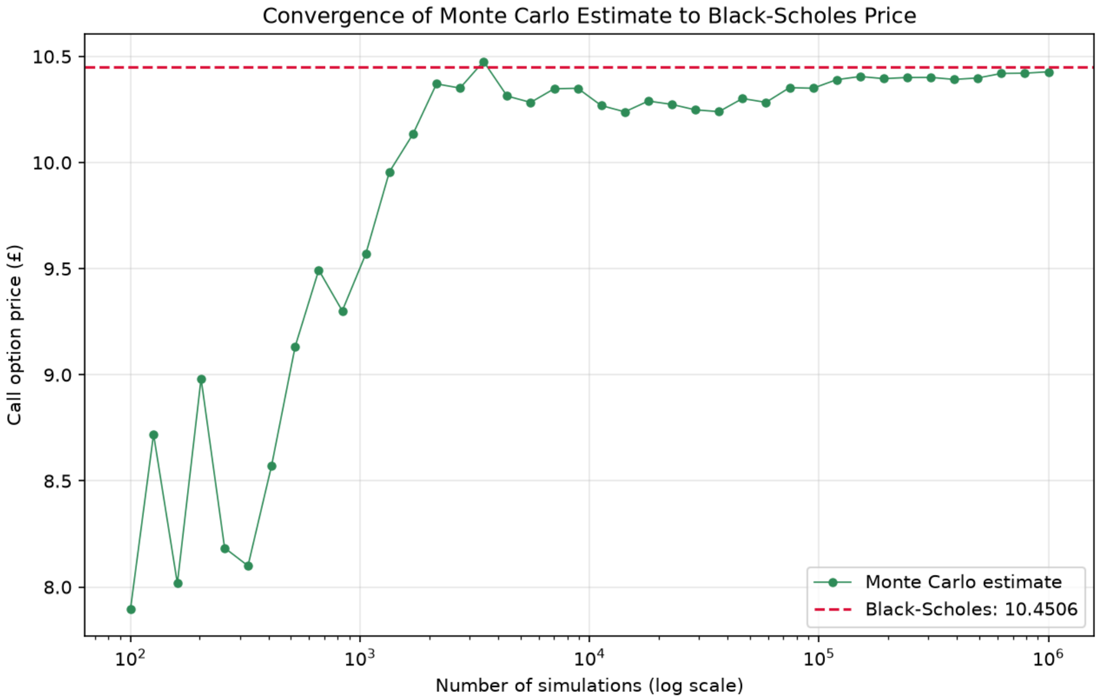

Figure 3: Monte Carlo price against the number of simulated paths n. The estimate settles onto the Black–Scholes price as the standard error shrinks at the $1/\sqrt{n}$ rate.

The method demonstrates an unbiased probabilistic estimator whose error is controlled by sample size rather than by a step length, converging slowly but in a way that does not depend on dimension.

6 The finite-difference method

The finite-difference method solves the Black–Scholes equation (2.4) directly, by replacing the derivatives with differences on a grid and marching backwards from the payoff. It is the method that generalises most readily to problems with no closed form, and it is also the one whose numerical behaviour, stability and convergence, most repays study, because both can be made precise and then checked against the exact price; a standard treatment of finite-difference schemes for parabolic equations of this kind is given in Ammari [4].

The scheme developed here is called *explicit* because each new value is computed directly from values that are already known, with no equations left to solve. The alternative, an *implicit* scheme, writes the new time level in terms of *other unknown* values at that same new time level, so every step requires solving a system of equations simultaneously rather than just substituting numbers in. Implicit schemes cost more per step for exactly that reason, but, as noted in section 8, they buy unconditional stability in return; the explicit scheme here is simpler to implement and to reason about, at the cost of the stability condition derived below.

6.1 Discretisation

Lay down a grid with prices $S_i = i\,\Delta S$ for $i = 0, \dots, M$ and times $t_n = n\,\Delta t$ for $n = 0, \dots, N$, and write V_i^n for the approximation to $V(S_i, t_n)$. The terminal condition fixes V_i^N as the payoff, and the boundary conditions at $S = 0$ and $S = S_M$ are known; the scheme fills in the interior by marching backwards in time.

The *explicit* scheme replaces every derivative in (2.4) with a difference between neighbouring grid values: each first derivative becomes a difference across adjacent nodes, and the second derivative a difference of those differences. Substituting these into (2.4) and rearranging for V_i^n gives the update rule used in the implementation, in which each value at one time level is a weighted combination of three values at the next:

$$V_i^n = a_i\, V_{i-1}^{n+1} + b_i\, V_i^{n+1} + c_i\, V_{i+1}^{n+1}, \tag{6.1}$$

with coefficients

$$a_i = \tfrac{1}{2}\Delta t\,(\sigma^2 i^2 - ri), \qquad b_i = 1 - \Delta t\,(\sigma^2 i^2 + r), \qquad c_i = \tfrac{1}{2}\Delta t\,(\sigma^2 i^2 + ri). \tag{6.2}$$

Here i is the price-step index, so $S_i = i\,\Delta S$. Each interior value at one time level is built from three values at the next, as in the stencil of fig. 4.

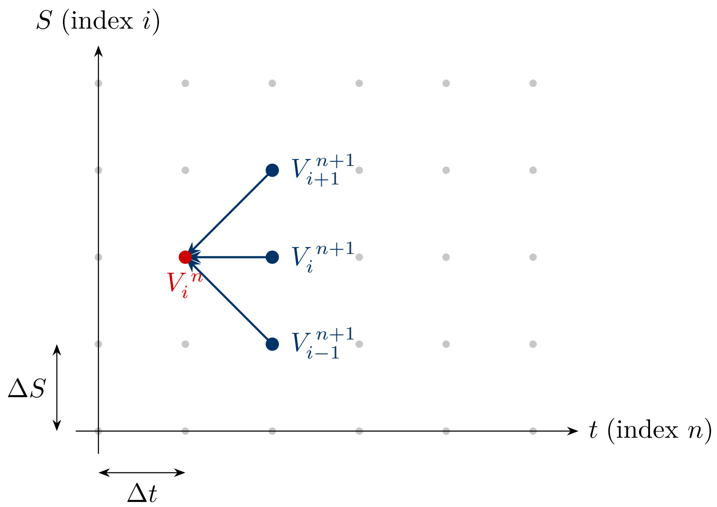

Figure 4: The explicit stencil. The value at an interior node (i, n) is a weighted combination of three nodes at the next time level, $(i-1, n+1)$, $(i, n+1)$ and $(i+1, n+1)$, with weights a_i, b_i and c_i from (6.2). The scheme marches backwards in time from the payoff.

6.2 Stability

The update (6.1) is a weighted combination of three values. If the weights are non-negative the combination is *monotone*: it behaves like a weighted average, and a weighted average can never fall outside the range of its ingredients, the way a mixture of three jugs of water can never end up colder than the coldest jug or hotter than the hottest. An error in the inputs is therefore blended, not magnified, and the scheme is stable. The off-diagonal weights a_i and c_i are non-negative for the relevant range of i, but the central weight

$$b_i = 1 - \Delta t\,(\sigma^2 i^2 + r)$$

can turn negative. This is the weight on V_i^{n+1}, the node directly to the right of the target node in fig. 4, one time step ahead. Reading the formula, b_i shrinks as i grows, because $\sigma^2 i^2$ grows with it, and shrinks further as Δt grows; once $\Delta t\,(\sigma^2 i^2 + r)$ exceeds 1, b_i flips negative. At that point the scheme is no longer a weighted average: it has to subtract a fraction of V_i^{n+1} rather than add one, so the monotone property above breaks, and the scheme becomes unstable. The binding case is the largest index $i = M$, where the coefficient $\sigma^2 i^2$ is greatest; requiring $b_M \geq 0$ gives $\Delta t \leq 1/(\sigma^2 M^2 + r)$, or, with $\Delta t = T/N$ and neglecting the small r term,

$$N \gtrsim \sigma^2 M^2 T. \tag{6.3}$$

Below this threshold b_M is negative, the scheme amplifies the highest-frequency components of the error at every step, and the solution diverges. (The same conclusion follows from von Neumann analysis, which tracks the growth of Fourier modes under the scheme; the positivity argument here is its elementary counterpart.)

Figure 5 shows both regimes at $M = 200$, where (6.3) requires $N \gtrsim \sigma^2 M^2 T = 1600$. With $N = 100$, far below the threshold, the scheme is unstable: the solution reaches a magnitude of around 10^{138} within a hundred time steps. With $N = 2000$, above the threshold, it is well behaved and returns a price of 10.4550.

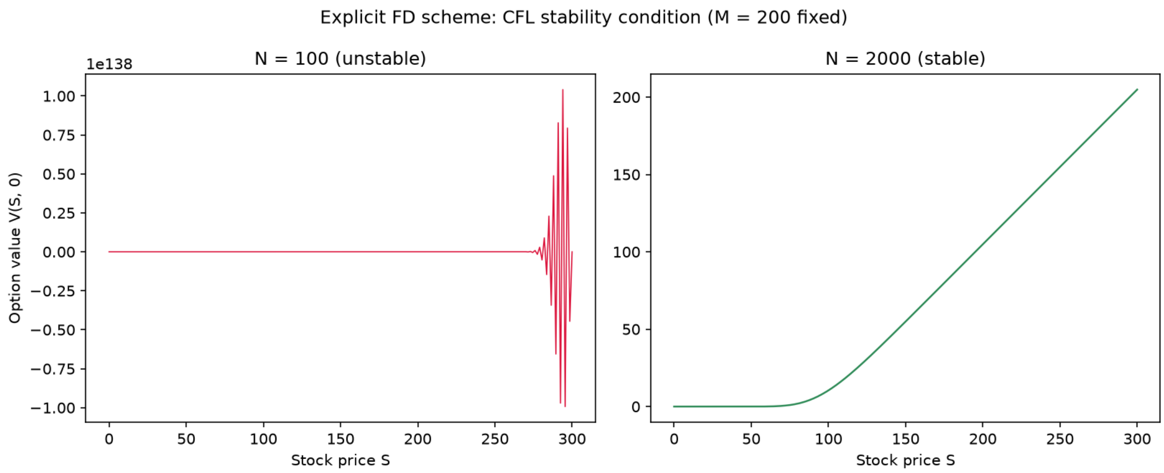

Figure 5: The explicit scheme at fixed $M = 200$ under an unstable and a stable choice of time step. With $N = 100$ the stability condition (6.3) is violated and the solution diverges (left); with $N = 2000$ it is satisfied and the scheme recovers the payoff structure and a correct price (right).

6.3 Convergence

With stability ensured, the question is how quickly the price approaches the exact answer as the grid is refined. The difference formulas used in section 6 only approximate the true derivatives they stand in for; *truncation error* is simply the name for the resulting gap, the small error left over in each approximation.

For the central differences used here, this gap is known to shrink with the square of the grid spacing ΔS, so halving ΔS should roughly quarter the error: a standard property of central differences, with the order-two space convergence of the explicit scheme set out in Ammari [4]. This is exactly what is observed. Figure 6 plots the error against the exact price on logarithmic axes as M increases. Logarithmic axes turn a relationship of the form error $\propto M^{\text{power}}$ into a straight line, whose slope is that power; here the points fall on a line of slope -2, meaning error $\propto 1/M^2$. That is exactly the quartering described above: doubling M multiplies the error by $2^{-2} = 1/4$, and the error does drop by a factor of about four each time M is doubled in the figure, confirming the quartering empirically.

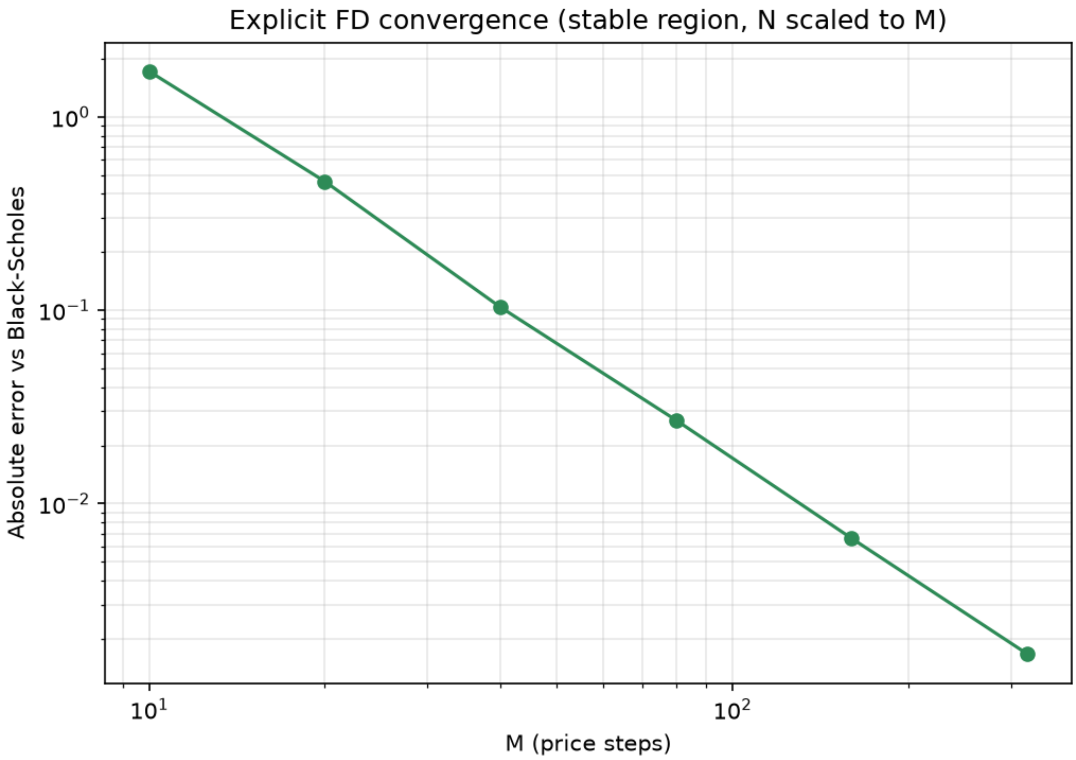

Figure 6: Error against the Black–Scholes price as the number of space steps M increases, on log–log axes. The slope of -2 (the error quartering each time M doubles) confirms second-order convergence in the space step.

The method demonstrates the discretisation of a partial differential equation, the conditional stability of the explicit scheme and how to diagnose it, and second-order convergence measured against a known answer. It is this controlled agreement that licenses applying the same machinery where no closed form exists.

7 American options and early exercise

An American option may be exercised at any time up to maturity, not only at T. The holder must decide, at every moment, whether to exercise now or to hold; the value is what an optimal exercise policy achieves. This is an optimal stopping problem, and it has no closed-form solution even under the present model. The binomial tree handles it with a single change.

7.1 Early exercise in the tree

At each node, compare the value of exercising immediately, the intrinsic value $(K-S)^+$ for a put, with the value of holding, which is the discounted risk-neutral expectation of the two child nodes. The node value is the larger of the two,

$$V = \max\Big((K-S)^+,\ e^{-r\Delta t}\big[\, p\, V_{\mathrm{up}} + (1-p)\, V_{\mathrm{down}} \,\big] \Big). \tag{7.1}$$

The only difference from the European recursion (4.3) is the outer maximum: one comparison at each node, asking whether exercising beats holding. The nodes where exercise wins form the exercise region; elsewhere it is better to hold. Figure 7 illustrates this for a put.

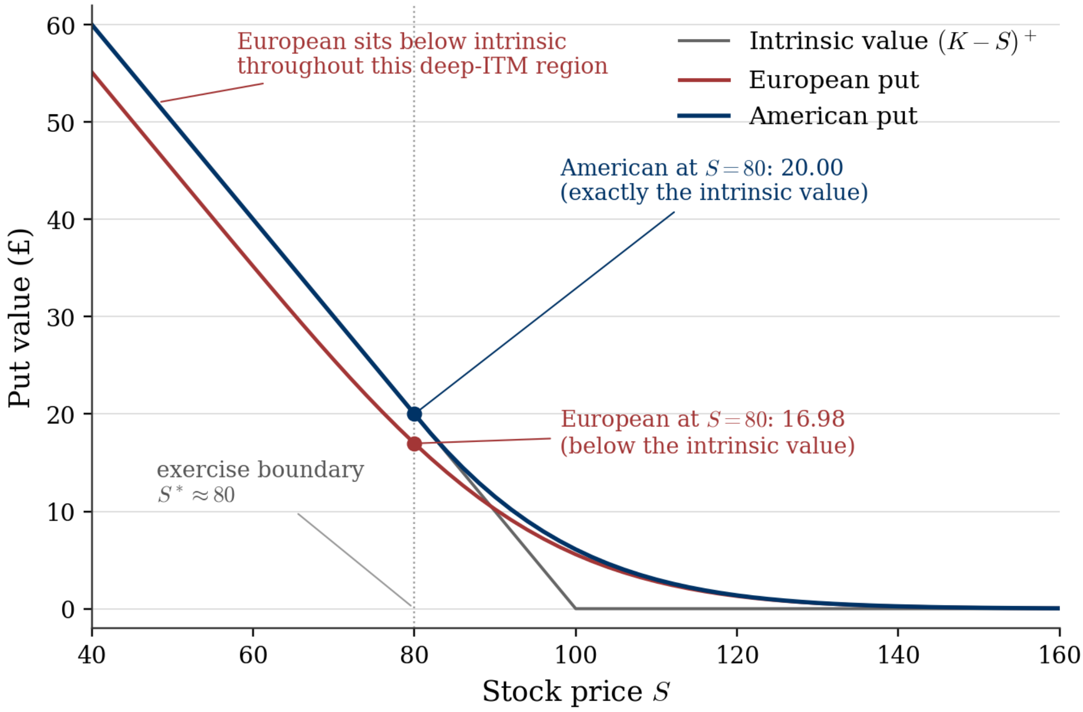

Figure 7: Put value against the stock price, at the benchmark parameters ($K = 100$, $r = 5\%$, $\sigma = 20\%$, $T = 1$): the intrinsic value, the European put (Black–Scholes), and the American put (binomial tree, $N = 400$). Below the exercise boundary S^* the American value coincides exactly with the intrinsic value; the European value sits below the intrinsic value throughout the deep in-the-money region, since a European holder cannot realise that value until T.

7.2 When early exercise matters

A comparison against intrinsic value alone is not enough to show what is distinctive about an American option. Both American and European options have *time value*: the chance that the stock moves favourably before T, so both are generally worth more than today's cash-in value while time remains; that is true of any option, not something special to the American contract. The feature that is specific to the American option is the early exercise *right*, and the only way to see its effect is to compare the American price against the European price for the same contract, which is what fig. 7 does.

Two features of that figure carry the argument. First, in the deep in-the-money region the European curve sits *below* the intrinsic value: a European holder is locked in and must wait until T to realise the payoff, so today's value is a discounted, and therefore smaller, version of it. Second, the American curve never does this: it cannot fall below intrinsic value, because the holder always has the right to exercise immediately and lock in that amount; arbitrage would otherwise be possible. The gap between the navy and red curves in that region, vanishing above S^*, is precisely the value of the early-exercise right, the American-specific feature, isolated from ordinary time value.

It is worth being clear about what is being compared here in a different sense too. Every method in this report assumes no dividends throughout: that is a standing simplification used everywhere, not something switched on or off in this section. Given that, the comparison above is specific to puts. For a call, by contrast, early exercise is never optimal under this no-dividend assumption: holding is always worth at least the intrinsic value, because exercising early forfeits both the interest earned by deferring payment of the strike and the remaining time value. The American call therefore equals the European call, with no early-exercise premium at all. (With dividends, a call can sometimes be worth exercising just before a payment, to capture it; that case does not arise here, since the model has none.)

Both predictions are borne out numerically. For a deep in-the-money put at $S = 80$, $K = 100$, the American price is exactly 20.0, the intrinsic value $K - S$, against a European put of 16.98: an early-exercise premium of about 3.0, visible in fig. 7 as the gap between the navy and red points at $S = 80$. For the call, the American and European prices agree to eleven decimal places. The test suite checks both: that the American put strictly exceeds the European put when deep in the money, and that the American call equals the European call with no dividends.

The method demonstrates the early-exercise premium and the optimal-stopping structure behind it. The tree accommodates early exercise almost for free, by the single maximum in (7.1); the other two methods do not, for reasons taken up next.

8 Conclusion

The four methods price the benchmark European call in close agreement, as table 2 records. They differ by amounts consistent with their respective numerical errors: the binomial and finite-difference values by small discretisation errors, the Monte Carlo value by a sampling error of the size its standard error predicts, which is evidence that the implementation is correct.

Method	Price	Difference from Black–Scholes
Black–Scholes (closed form)	10.4506	-
Binomial tree (CRR, $N = 5000$)	10.4502	0.0004
Monte Carlo (10^6 paths)	10.4532 ± 0.0147	0.0026
Explicit finite difference ($M = 200$, $N = 2000$)	10.4550	0.0044

Table 2: The four methods at $S = K = 100$, $r = 5\%$, $\sigma = 20\%$, $T = 1$. The numerical methods agree with the exact price to within their expected errors.

Each method also shows something the others do not. The closed form is exact and fixes the reference. The binomial tree is a fully discrete scheme that converges to the continuous price, with a visible odd–even oscillation along the way. Monte Carlo is an unbiased estimator whose error is set by sample size and falls only as $1/\sqrt{n}$, but which is indifferent to dimension. The finite-difference scheme solves the governing equation directly and brings the sharpest numerical questions with it: it is stable only when the time step is fine enough to keep the central coefficient (6.2) non-negative, and it converges at second order in the space step, both confirmed here against the exact price. The American extension adds the early-exercise premium and the optimal stopping that produces it.

The scope was kept narrow on purpose, and the limitations follow from it. Volatility and the interest rate are constant; the explicit scheme is only conditionally stable; and the Monte Carlo and finite-difference treatments here price European options. Three extensions follow naturally.

First, American pricing beyond the tree. In the tree early exercise is a single comparison because each node carries its own continuation value. Monte Carlo does not: a simulated path knows only its own future, not the continuation value at an interior point, so that value must be estimated by regressing discounted future payoffs on the current state, as in the Longstaff–Schwartz method [3]. The finite-difference scheme must instead enforce the exercise constraint at every grid point, which turns the equation into a linear complementarity problem, solved by projected successive over-relaxation or a penalty method.

Second, unconditional stability. The explicit scheme's condition (6.3) forces the number of time steps to grow with the square of the space refinement. An implicit scheme, or the Crank–Nicolson scheme, is unconditionally stable and removes that constraint, at the cost of solving a linear system at each step.

Third, beyond constant coefficients. Local or stochastic volatility, or a term structure of interest rates, would be the next step towards a model that bears on traded prices, and each of the methods developed here extends, with more work, to accommodate it.

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Further reading. Readers seeking a fuller, textbook-length treatment of this material may consult the standard references: P. Wilmott, *Paul Wilmott on Quantitative Finance*, 2nd edition, Wiley, 2006; P. Glasserman, *Monte Carlo Methods in Financial Engineering*, Springer, 2003; and D. J. Higham, *An Introduction to Financial Option Valuation*, Cambridge University Press, 2004.